

A
Mini project report
On
SMART LAWN SPRINKLER AUTOMATIC SYSTEM

Submitted in partial fulfilment of the Requirements for the award of Degree of
BACHELOR OF TECHNOLOGY

in
Electronics & Communication Engineering
BY

A. Johnson (23681A0404)

Under the esteemed guidance of

Mr. D. Jagan

Assistant Professor



Dept. Of Electronics & Communication Engineering
CHRISTU JYOTHI INSTITUTE OF TECHNOLOGY & SCIENCE
(Affiliated to JNTU, Hyderabad)

Colombo Nagar, Yeshwanthapur, Jangaon-506167, Telangana

2025-2026

CHRISTU JYOTHI INSTITUTE OF TECHNOLOGY & SCIENCE

(Affiliated to JNTU, Hyderabad)

Colombo Nagar, Yeshwanthapur, Jangaon-506167, Telangana

2025-2026

Dept. of Electronics & Communication Engineering



Certificate

This is to certify that that the report entitled “Smart Lawn Sprinkler Automatic System” that is being submitted by **A.Johnson (23681A0404)** in partial fulfillment for the requirement for the award of the Degree of Bachelor of Technology in “**Electronics & Communication Engineering**” and submitted to **Christu Jyothi Institute of Technology and Science, Jangaon**, is a record of Bonafide work carried out by them under our guidance and supervision.

Signature of Guide

Mr. D. Jagan

Signature of HOD

Mr. Allanki Sanyasi Rao

Signature of External examiner

CHRISTU JYOTHI INSTITUTE OF TECHNOLOGY & SCIENCE

(Affiliated to JNTU, Hyderabad)

Colombo Nagar, Yeshwanthapur, Jangaon-506167, Telangana

2025-2026

Dept. of Electronics & Communication Engineering



Institution Vision and Mission

VISION

To admit and groom students from rural background and be a truly rural technical institution benefiting society and nation as a whole institute.

MISSION

- The mission of the institution is to create, deliver and refine knowledge. Being a rural technical institute, our mission is to.
- Enhance our position to one of the best technical institutions and to measure our performance against the highest defined standards.
- Provide highest quality learning environment to our students for their greater well-being so as to equip them with highest technical and professional ethics.
- Produce engineering graduates fully equipped to meet the ever-growing needs of industry and society



Principal
Christu Jyothi Institute of Technology & Science
Colombo Nagar, Yeshwanthapuram (Vill)
Jangaon (Mdl), Jangaon (Dist)-506167.

PRINCIPAL

CHRISTU JYOTHI INSTITUTE OF TECHNOLOGY & SCIENCE

(Affiliated to JNTU, Hyderabad)

Colombo Nagar, Yeshwanthapur, Jangaon-506167, Telangana

2025-2026

Dept. of Electronics & Communication Engineering



Department Vision and Mission

VISION

To be an established center of excellence in Electronics and Communication Engineering facilitating youth towards professional, leadership and industrial needs.

MISSION

- Impart theoretical and practical technical education of high standard with quality resources and collaborations.
- Organize trainings and activities towards Overall personality development in time with industrial need.
- Promote innovation towards sustainable solutions with multi discipline team work with ethics.

HOD

DECLARATION

We here by declare that the document entitled “**Smart Lawn Sprinkler Automatic System**”submitted to the **Christu Jyothi Institute of Technology and Science** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (B-Tech) in Electronics and communication engineering is a record of an original work done by under the guidance of **Mr. D. Jagan Assistant Professor** and this document has not been submitted to any other university for the award of any other.

PROJECT MEMBER:

A. Johnson (23681A0404)

ACKNOWLEDGEMENT

We here by express our sincere gratitude to the **Management of Christu Jyothi Institute of Technology & Science** for their kind encouragement bestowed up on us to do this Mini project.

We earnestly take the responsibility to acknowledge the following distinguished personalities who graciously allowed our project work successfully.

We express our sincere thanks to our director **Rev.Fr. D. Vijaya Paul Reddy**, Principal **Mr. Dr. S. Chandrashekhar Reddy** for their encouragement, which has motivated us to strive hard to excel in our discipline of engineering.

We are greatly indebted to the Head of the Department **Mr. Allanki Sanyasi Rao, Assistant Professor** for his motivation and the project guide **Mr. D. Jagan, Assistant Professor**, He has been responsible for providing us with lot of splendid opportunities, which has shaped our career. His advices, ideas and constant support has engaged us and helped us to get through in difficult times. His excellent guidance has made the timely completion of this Mini Project.

Our heartfelt thanks to our Project Coordinator, **Mr. K Chiranjeevi, Assistant Professor** for the support and advice for our innovative project he has given us through our project reviews. We also wish to thank them for their guidance and support during our early days in the area of lot.

Last but not the least, we express our gratitude to the Teaching and Nonteaching Staff of the Department of Electronics and Communication for their needy and continuous support in technical assistance.

ABSTRACT

Smart lawn sprinkler automatic system is a mini-project aimed at developing an intelligent irrigation. solution that automates lawn watering based on real-time environmental conditions and soil moisture levels. The system uses soil moisture sensors to continuously monitor moisture content in the ground and a microcontroller to process these readings. When the moisture falls below a predetermined threshold, the system automatically activates the sprinkler via a relay or actuator, ensuring the lawn receives water only when necessary

This automation eliminates the need for manual watering, reduces water wastage, and improves lawn health and maintenance efficiency. The project can be extended with IoT functionality to allow remote monitoring and control via a mobile app, enabling users to view soil conditions and adjust watering schedules from any location. The smart irrigation approach supports sustainable water usage and reduces labour effort compared to conventional manual. Water conservation and efficient irrigation management are critical challenges in modern landscaping and agriculture.

The Smart Lawn Sprinkler Automatic System is designed to optimize water usage by automatically controlling irrigation based on real-time environmental conditions. The system utilizes soil moisture sensors, temperature and humidity sensors, and weather-based inputs to determine the precise watering requirements of the lawn. A microcontroller processes the sensor data and activates the sprinkler only when the soil moisture falls below a predefined threshold, thereby preventing overwatering.

TABLE OF CONTENTS

CHAPTER.NO	TITLE	PAGE.No
1	Introduction	1
	1.1 Overview	
	1.2 Methodology	2
2	Literature Review	3
	2.1 Problem statement	4
	2.2 Proposed solution	5
	2.3 Existing solution	6
3	Design	7-9
	3.1 Block Diagram	7
	3.2 Schematic Diagram	8
	3.3 Description	9
4	Hardware Components	10-17
	4.1 List of Components	10
	4.2 Operation of components	11-16
5	Working Principle	18-19
6	Result	20-21
7	Advantages & Dis-Advantages	22-24
	7.1 Advantages	22
	7.2 Dis-Advantages	23
	7.3 Application	24
8	Conclusion & Future Scope	25-26
	8.1 Conclusion	25
	8.2 Future scope	26
	Reference	27
	Appendix	28-31

LIST OF FIGURES

FIGURE. NO	FIGURE	PAGE. NO
3.1	Block Diagram	7
3.2	Schematic Diagram	8
4.2.1	ESP-8266	11
4.2.2	Servo motor	12
4.2.3	Water Pump	13
4.2.4	Relay module (Dual)	14
4.2.5	Soil moisture sensor	15
4.2.6	Power supply	16